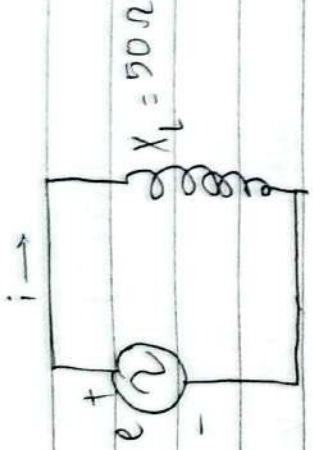


Tutorial - 5

1)



Soln:

$$e = 100 \sin(157t + 30^\circ) = \frac{100}{\sqrt{2}} \angle 30^\circ$$

Also,

$$Z = 0 + j50 = 50 \angle 90^\circ$$

$$\therefore I = \frac{E}{Z} = \frac{100/\sqrt{2} \angle 30^\circ}{50 \angle 90^\circ} = \sqrt{2} \angle -60^\circ$$

$$\therefore i = \sqrt{2} \times \sqrt{2} \sin(157t - 60^\circ) = 2 \sin(157t - 60^\circ)$$

$$\therefore X_L = \omega L = 50$$

Also,

$$\omega = 157$$

$$\therefore L = \frac{50}{157} = 318.47 \text{ mH}$$

$$\text{Average Power} = \frac{V_m I_m \cos \theta}{2} = \frac{V_m I_m \cos 90^\circ}{2} = 0 \text{ W}$$

(2) Soln:

power factor, $\cos \theta = 0.5$ lagging

$$\therefore \theta = \cos^{-1} 0.5$$
$$= -60^\circ$$

$$\text{Input voltage, } u = 50 \sin(\omega t + 10^\circ) = \frac{50}{\sqrt{2}} \text{ V } \angle 10^\circ$$

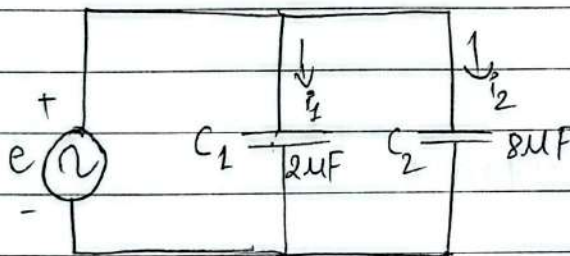
$$\text{power} = V_{\text{rms}} \cdot I_{\text{rms}} \cos \theta = 500 \text{ W}$$

$$\text{or, } I_{\text{rms}} = \frac{500}{\frac{50 \times 0.5}{\sqrt{2}}} = 20\sqrt{2} \text{ A}$$

$$\text{or, } I_m = \sqrt{2} \times 20\sqrt{2} = 40 \text{ A}$$

$$\therefore i = 40 \sin(\omega t + (10^\circ - 60^\circ)) = 40 \sin(\omega t - 50^\circ)$$

(3)



$$e = \sqrt{2} 100 \sin(10^4 t + 60^\circ)$$
$$= 100 \angle 60^\circ$$

$$Z_1 = 0 - j \times C_1 = -j \frac{1}{\omega C} = -j \frac{1}{10^4 \times 2 \times 10^{-6}} = -j 50 = 50 \angle -90^\circ$$

$$Z_2 = 0 - j \times C_2 = -j \frac{1}{10^4 \times 8 \times 10^{-6}} = -j 12.5 = 12.5 \angle -90^\circ$$

$$Z_3 = 0 - j \times C = -j \frac{1}{10^4 \times 10 \times 10^{-6}} = -j 10 = 10 \angle -90^\circ$$

$$i_1 = \frac{e}{Z_1} = \frac{100 \angle 60^\circ}{50 \angle -90^\circ} = 2 \angle 150^\circ = 2.828 \sin(10^4 t + 150^\circ)$$

$$i_2 = \frac{e}{Z_2} = \frac{100 \angle 60^\circ}{12.5 \angle -90^\circ} = 8 \angle 150^\circ = 11.313 \sin(10^4 t + 150^\circ)$$

$$i_s = \frac{e}{Z} = \frac{100 \angle 60^\circ}{10 \angle -90^\circ} = 10 \angle 150^\circ = 14.14 \sin(10^4 t + 150^\circ)$$

4) Soln:

$$R = 100 \Omega, C = 50 \mu F, f = 50 \text{ Hz}$$

$$\therefore X_C = \frac{1}{2\pi \times 50 \times 50 \times 10^{-6}} = 63.662 \Omega$$

$$\therefore Z = 100 - j63.662 = 118.544 \angle 32.48^\circ \Omega$$

$$I = \frac{200}{118.544} = 1.687 \text{ A}$$

$$p.f. = \cos(-32.48^\circ) = 0.8435$$

$$\text{phase angle} = 32.48^\circ$$

$$\text{voltage across resistor} = I \times R = 1.687 \times 100 = 168.7 \text{ V}$$

$$\text{voltage across capacitor} = I \times X_C = 1.687 \times 63.662 = 107.39$$

(5) soln:
 $Z_1 = (5 + j7) \Omega = 8.6 \angle 54.46^\circ$

$$Z_2 = (10 - j5) \Omega = 11.18 \angle 26.56^\circ$$

$$Z_T = Z_1 + Z_2 = 5 + j7 + 10 - j5 = 15 + j2 = 15.13 \angle 7.59^\circ$$

$$\therefore I = \frac{200}{15.13} = 13.218 \text{ A}$$

$$\text{active power} = 200 \times 13.218 \cos 7.59^\circ \\ = 2620 \text{ W}$$

$$\text{apparent power} = 2643.6 \text{ VA}$$

$$\text{power factor} = \cos(-7.59^\circ) = 0.991 \text{ lagging}$$

(6) soln:

$$R = 200 \Omega, f = 50 \text{ MHz}$$

$$C = 10 \text{ pF}, X_C = \frac{1}{2\pi \times 50 \times 10^6 \times 10 \times 10^{-12}} = 318.3 \Omega$$

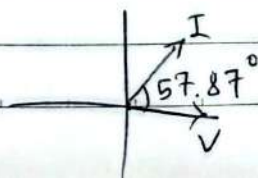
$$\therefore Z = 200 - j318.3 = 375.918 \angle -57.857^\circ$$

$$I = \frac{12}{375.918} = 31.92 \text{ mA}$$

$$p.f. = \cos(-57.857^\circ) = 0.532 \text{ leading}$$

$$\text{active power} = 12 \times 31.92 \times 10^{-3} \times 0.532 = 0.20379 \text{ W}$$

phasor diagram is as shown below



7) $R = 10 \Omega$, $L = 0.1 H$, $C = 150 \mu F$, $V = 200 V$ and $f = 50 Hz$

$$X_L = 2\pi f L = 31.41 \Omega$$

$$X_C = \frac{1}{2\pi f C} = 21.22 \Omega$$

$$Z = 10 + j 31.41 - j 21.22 = 10 + j 10.19 = 14.277 \angle 45.53^\circ$$

$$I = \frac{V}{Z} = \frac{200}{14.277} = 14 A$$

$$pf = \cos(-45.53^\circ) = 0.7 \text{ lagging}$$

$$\text{voltage across capacitor} = I \times X_C = 297 V$$

(8) $R = 68 \Omega$, $L = 10 mH$, $V = 33 V$, $f = 1 kHz$

$$X_L = \omega L = 2\pi f L = 62.83 \Omega$$

$$Z = 68 + j 62.83 = 92.58 \angle 42.737^\circ$$

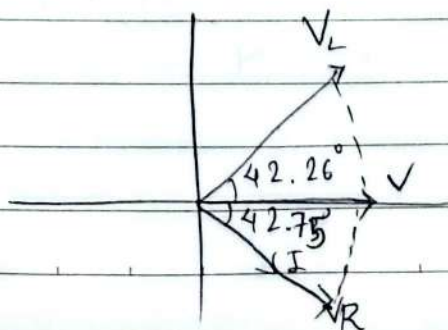
$$I = \frac{33}{92.58} = 0.356 A \angle -42.737^\circ$$

$$\text{Voltage across resistor} = 0.356 \times 68 = 24.2 V \angle -42.75^\circ$$

$$\text{Voltage across inductor} = 22.39 V \angle 42.26^\circ$$

$$\text{phase angle} = 42.737^\circ$$

The phasor diagram is as shown below,



(9) $R = 20 \text{ k}\Omega$, $L = 1 \text{ H}$, $C = 0.005 \text{ F}$, $V = 100 \text{ V}$, $f = 1000 \text{ Hz}$

$$X_L = 2\pi fL = 6283.18$$

$$X_C = \frac{1}{2\pi fC} = 0.03183$$

$$\therefore Z = 20 \times 10^3 + j6283.18 - j0.03183 = 20000 + j6283.18 = 20963.7 \angle 17.44^\circ$$

$$I = \frac{100}{20963.7} = 4.77 \text{ mA} \angle -17.44^\circ$$

$$\text{phase angle} = 17.44^\circ$$

$$p.f. = 0.954 \text{ lagging}$$

$$\text{active power} = 100 \times 4.77 \times 10^{-3} \times 0.954 = 0.455 \text{ W}$$

(10) $R = 100 \Omega$, $L = 0.6 \text{ H}$, $V = 230 \text{ V}$, $f = 50 \text{ Hz}$

$$X_L = 2\pi fL = 188.49 \Omega$$

$$Z = \frac{100 + j188.49}{100 + j188.49} = \frac{18849 \angle 90^\circ}{213.37 \angle 62.05^\circ} = 88.33 \angle 27.95^\circ$$

$$I = \frac{230 \angle 0^\circ}{88.33 \angle 27.95^\circ} = 2.6 \angle -27.95^\circ$$

$$\phi = +27.95^\circ$$

$$\text{power factor} = 0.88 \text{ lagging}$$

$$\text{active power} = 528.24 \text{ W}$$

(11) $R = 500 \Omega$, $C = 18 \mu F$, $V = 250 V$, $f = 50 Hz$

$$X_C = \frac{1}{2\pi f C} = 176.838 \Omega$$

$$Z = \frac{500 \times (-j176.838)}{500 - j176.838} = \frac{88419 \angle -90^\circ}{530.35 \angle -19.477^\circ} = 166.718 \angle -70.52^\circ$$

$$I = \frac{250 \angle 0^\circ}{166.718 \angle -70.52^\circ} = 1.499 \angle 70.52^\circ$$

$$\phi = 70.52$$

$$\text{active power} = 250 \times 1.499 \cos 70.52^\circ = 124.97 W$$

(12) $L = 0.6 H$, $R = 100 \Omega$, $C = 30 \mu F$, $V = 230 V$, $f = 50 Hz$

$$X_L = 2\pi f L = 188.49 \Omega, \quad X_C = \frac{1}{2\pi f C} = 106.1 \Omega$$

$$\frac{1}{Z} = \frac{1}{100} + \frac{1}{188.49 \angle 90^\circ} + \frac{1}{106.1 \angle -90^\circ}$$

$$= 0.01 + \frac{1 \angle -90^\circ}{188.49} + \frac{1}{106.1 \angle -90^\circ}$$

$$= 0.01 - j \frac{1}{188.49} + j \frac{1}{106.1} = 0.01 + j 0.004119$$

$$= 0.0108 \angle 22.386^\circ$$

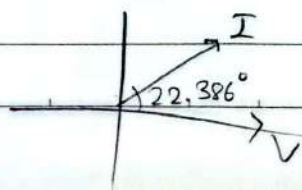
$$\therefore Z = 92.59 \angle -22.386^\circ$$

$$\therefore I = \frac{230 \angle 0^\circ}{92.59 \angle -22.386^\circ} = 2.484 A \angle 22.386^\circ$$

$$\text{phase angle} = 22.386^\circ$$

$$\text{active power} = 230 \times 2.484 \times \cos 22.386^\circ = 528.26 W$$

The phasor diagram is as shown below:



$$(13) Z_1 = 4 - j10 = 10.77 \angle -68.198^\circ \Omega$$

$$Z_2 = 6 + j8 = 10 \angle 53.13^\circ \Omega$$

$$I_1 = \frac{230 \angle 0^\circ}{10.77 \angle -68.198^\circ} = 21.355 A \angle 68.198^\circ = 7.931 + j19.824$$

$$I_2 = \frac{230 \angle 0^\circ}{10 \angle 53.13^\circ} = 23 A \angle -53.13^\circ = 13.8 - j18.399$$

$$I = I_1 + I_2 = 21.77 \angle 3.759^\circ$$

$$Z = \frac{V}{I} = \frac{230 \angle 0^\circ}{21.77 \angle 3.759^\circ} = 10.565 \angle -3.759^\circ$$

$$p.f. = \cos(3.759^\circ) = 0.9978 \text{ leading}$$

$$\text{active power} = 230 \times 21.77 \times 0.9978 = 4996.08 \text{ W}$$

$$\text{reactive power} = 230 \times 21.77 \times \sin(3.759^\circ) = 328.26 \text{ VAR}$$

$$\text{apparent power} = 230 \times 21.77 = 5007.1 \text{ VA}$$

The phasor diagram is as shown

